

XINYUAN LIU

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Education

Arizona State University

Ph.D. in Computer Science

Tempe, Arizona

Aug. 2025 – Present

- Advisor: Ransalu Senanayake
- GPA: 4.0/4.0

Rice University

M.S. in Computer Science

Houston, Texas

Aug. 2022 – Dec. 2023

- GPA: 3.87/4.0

Huazhong University of Science & Technology

B.S. in Software Engineering

Wuhan, China

Sept. 2018 – Jun. 2022

- GPA: 3.79/4.0

Research Interests

Information-efficient AI systems. I study how intelligent systems can make better decisions from limited but informative signals. My work connects LLM evaluation and routing, cognitive evaluation for humanoid robots, imitation learning for embodied agents, and multimodal learning for surgical and medical systems.

Preprints & Publications

1. Isaiah Milkey, Som Sagar, Aditya Taparia, **Xinyuan Liu**, Jiqing Wen, Ransalu Senanayake. PhysicsLENS: Diagnosing Physical Property Blindness in Video Generation Models. under review, 2026.
2. Eren Sadikoglu, Aditya, **Xinyuan Liu**, Ransalu Senanayake. ROOT: Discovering Rewards for User-Specified Embodied Behaviors. under review, 2026.
3. Son Nguyen, **Xinyuan Liu**, Ransalu Senanayake. CUPID in the Model Zoo: Online Matchmaking for Selecting Your Dream LLM. *International Conference on Machine Learning (ICML)*, 2026.
4. Ashkan Shahbazi, Kyvia Pereira, Jon S. Heiselman, Elaheh Akbari, Annie Benson, Sepehr Seifi, **Xinyuan Liu**, Garrison L. H. Johnston, Jie Ying Wu, Nabil Simaan, Michael Miga, Soheil Kolouri. Neural-Augmented Kelvinlet for Real-Time Soft Tissue Deformation Modeling. *AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
5. **Xinyuan Liu**, Eren Sadikoglu, Riana Chatterjee, Ransalu Senanayake. Physical Agentic AI: An Architecture for Orchestrating a Robot Crew with LLMs. *arXiv:2608.22657*, 2026. Abridged version in *IROS 2026 Workshop: Foundation Models in Multi-Robot Systems*.
6. **Xinyuan Liu**, Eren Sadikoglu, Ransalu Senanayake, Lixiao Huang. Cognitive Models as Evaluation Primitives in Humanoid Factors. *arXiv:2602.10069*, 2026. Abridged version in *ICLR 2026 Workshop on From Human Cognition to AI Reasoning*.
7. Son Nguyen, **Xinyuan Liu**, Ransalu Senanayake. LLM Routing as Reasoning: A MaxSAT View. *ICLR 2026 Workshop on Logical Reasoning of Large Language Models*, 2026.

Experience	Research Assistant	Aug. 2025 – Present
	<i>Arizona State University, advised by Dr. Ransalu Senanayake</i>	
	• Design online LLM selection and routing methods that identify informative user–model interactions, combining bandit learning, user studies, and large-scale evaluation to personalize model choice under limited feedback. • Build humanoid robotics pipelines on Unitree G1 and Go2 platforms for tele-operation, demonstration collection, behavior cloning, and policy evaluation, with an emphasis on data-efficient skill acquisition.	
	Research Assistant	Apr. 2024 – Apr. 2025
	<i>Vanderbilt University, advised by Dr. Jie Ying Wu</i>	
	• Built a GPT-style multimodal foundation model that fuses surgical video and kinematics to forecast robot end-effector trajectories, achieving < 5 mm RMSE for 1-second-ahead predictions and strong cross-task generalization. • Developed an AR surgical-training simulator that integrates video, kinematics, and gaze with ResNet, TCN, LSTM, and GCN models, improving gesture recognition and skill assessment by 18%.	
	Software Engineer Intern	May 2023 – Aug. 2023
	<i>Ra Labs, Jersey City, NJ</i>	
	• Fine-tuned GPT-2, Falcon, RWKV, and CodeT5 models for text-to-SQL, using QLoRA quantization to reduce memory footprint and support efficient adaptation. • Built a reproducible benchmarking framework with task-specific metrics to compare model outputs across datasets and identify reliable, data-efficient adaptation strategies.	
	Research Assistant	Sept. 2021 – Jun. 2022
	<i>Huazhong University of Science & Technology, advised by Dr. Shuguang Su</i>	
	• Integrated BERT and ResNet-50 to combine clinical text and chest imaging for lung disease diagnosis, achieving a micro-F1 score of 0.91 and reducing diagnosis time by 90%. • Stress-tested multimodal robustness under textual perturbations to identify failure modes and maintain stable diagnostic performance.	
Service	Reviewer: IROS, NeurIPS, CoRL	2026
	Workshop Reviewer: IROS, NeurIPS	2026
Skills	Programming Languages: Python, C#, C, C++, MATLAB, JavaScript, Java Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Hugging Face Transformers, OpenCV LLM & Agentic AI Tools: LlamaIndex, LangChain, OpenAI API, prompt engineering, retrieval-augmented generation, LLM evaluation Robotics & Simulation: ROS2, MuJoCo, Isaac Gym/Sim/Lab, Unity, robot learning, imitation learning Software & Research Engineering: Docker, FastAPI, Git, CI/CD, Linux, L ^A T _E X, experiment tracking Databases & Cloud Services: MySQL, MongoDB, AWS	